RESPONSIBLE WEB-SCALE DATA CURATION

Technical Report | Common Crawl WET Sample | Apache Spark and Toxic-BERT

Abstract. This report documents an implemented data-curation workflow using a 10,000-document Common Crawl WET sample. The pipeline performs Spark ingestion, exact deduplication, rule-based quality filtering, PII pattern redaction, Toxic-BERT screening, and descriptive language-wise analysis. After processing, 6,879 documents were retained. Toxicity screening flagged 117 of 6,996 documents (1.67%) at a threshold of 0.50. Language-wise removal rates differed in the observed sample, but the available evidence does not establish model bias. The experiment demonstrates an end-to-end prototype while exposing the computational cost of transformer inference and the need for stronger fairness evaluation.

1. Objectives and System Overview

Web-scale text collections can contain duplicate, low-quality, sensitive, and potentially toxic material. The objective is to prepare a higher-quality, privacy-aware corpus for downstream machine learning while measuring processing performance and documenting responsible-AI limitations.

Implemented stages: (1) WET extraction and ingestion; (2) exact deduplication using normalized text and SHA-256 hashes; (3) quality filtering; (4) PII pattern detection and redaction; (5) Toxic-BERT scoring and threshold-based flagging; and (6) language-wise descriptive analysis using automatic language identification.

Spark ran in local mode with two worker threads (local[2]). Toxicity inference used CPU execution, batches of eight, and a maximum input length of 512 tokens. The toxicity stage supports JSONL checkpointing so completed predictions can be reused after interruption.

2. Dataset and Experimental Method

The input was a WET segment from Common Crawl collection **CC-MAIN-2026-43**. Up to 10,000 conversion records were extracted. Each record contained a document identifier, domain, crawl date, and text. Extracted data was stored as JSONL and intermediate Spark datasets as Parquet.

Attribute	Value
Source	Common Crawl
Collection	CC-MAIN-2026-43
Format	WET archive (.warc.wet.gz)
Sample size	10,000 documents
Intermediate formats	JSONL and Apache Parquet
Processing framework	Apache Spark 3.5.6 / PySpark

The quality filter removed empty text, documents shorter than 200 characters, documents with fewer than 30 words, and documents whose ASCII alphabetic-character ratio was below 50%. PII detection used regular expressions for email- and phone-like strings. Documents containing detected patterns were redacted rather than removed.

3. Pipeline Results

The following document counts are based on recorded execution metrics. PII redaction preserves record count because it replaces detected values rather than excluding documents.

Stage	Input	Output	Observed change
Ingestion	—	10,000	Sample loaded
Deduplication	10,000	9,833	167 removed (1.67%)
Quality filtering	9,833	6,996	2,837 removed (28.85%)
PII redaction	6,996	6,996	0 records removed
Toxicity screening	6,996	6,879 retained	117 flagged (1.67%)

3.1 Deduplication

Text was normalized by lowercasing, trimming, and collapsing whitespace before SHA-256 hashing. The run identified 42 duplicate groups and removed 167 duplicate copies, leaving 9,833 unique documents. This detects exact duplicates after normalization, not semantic or near-duplicate documents.

3.2 Quality Filtering

Removal reason	Documents removed
Empty documents	0
Short documents	335
Low word count	35
Low ASCII alphabetic ratio	2,467
Total	2,837

The ASCII alphabetic-ratio condition accounted for 2,467 of 2,837 quality-filter removals (approximately 86.96%). This rule may disadvantage content written in non-Latin scripts. This is a potential design limitation, not a causal estimate of language discrimination.

3.3 PII Detection and Redaction

The PII stage processed 6,996 documents. Email-like patterns were detected in 2,046 documents and phone-like patterns in 1,206. Counts can overlap when a document contains both. All 6,996 records were retained after detected values were replaced with redaction placeholders. Regular expressions cannot guarantee detection of every personal-data instance.

3.4 Toxicity Screening

The pretrained *unitary/toxic-bert* model scored each of the 6,996 PII-redacted documents. Scores at or above 0.50 were flagged as potentially toxic. The model flagged 117 documents and retained 6,879 (1.67% flagged; 98.33% retained). Mean toxicity score was 0.04203.

Toxicity metric	Measured value
Input documents	6,996
Flagged documents	117
Retained documents	6,879
Threshold	0.50
Mean score	0.04203
Execution time	3,019.96 seconds (50.33 minutes)
Throughput	2.32 documents/second

Interpretation: Toxicity predictions are screening signals, not verified ground-truth labels. No full-corpus manual annotation was available to calculate reliable precision, recall, false-positive rate, or false-negative rate.

4. Language-wise Observations and Fairness

The analysis used *langdetect* to assign automatic language labels to toxicity-scored documents and calculate observed flag/removal rates by detected language. This comparison is descriptive: language identification may be imperfect, and groups may contain different topics, content types, and sample sizes.

Detected language	Documents	Flagged	Observed removal rate
Vietnamese (vi)	125	36	28.80%
English (en)	3,899	46	1.18%

The observed difference warrants further investigation but does not establish that the model is biased against Vietnamese. Content distribution, language-identification errors, and model quality across languages are possible contributing factors. Groups with fewer than 30 documents were excluded from the notebook's comparison chart.

Topic-wise fairness analysis remains pending because the extracted corpus does not contain validated topic labels. The project also lacks manually verified dialect or language-variety labels, so it cannot make claims about Indian English or other dialects. A stronger evaluation requires a stratified, manually annotated sample with topic, language variety, and toxicity labels.

5. Performance and Runtime

Recorded stage-level execution metrics show that transformer-based toxicity screening dominates runtime. The measurements below are experiment timings, not a distributed-cluster scalability benchmark.

Stage	Time (seconds)	Throughput (docs/sec)
Ingestion	5.73	1,745.99
Deduplication	11.73	852.63
Quality filtering	15.09	651.57
PII redaction	12.56	556.89
Toxicity filtering	3,019.96	2.32
Total measured stage time	3,065.67 (51.08 min)	—

Toxicity inference used CPU, batch size eight, and a maximum sequence length of 512 tokens. The long runtime illustrates the trade-off between transformer-based screening and throughput. Wall-clock measurements depend on hardware and runtime conditions. The experiment does not establish performance on a distributed cluster or at larger corpus sizes.

6. Testing and Reproducibility

The integrated unit-test suite completed with 20 passing tests and no failures. Tests cover topic fairness metric calculations, language fairness metric calculations, PII pattern detection/redaction, and toxicity threshold/filtering behaviour. These tests validate component logic; they are not a manually labelled evaluation of Toxic-BERT on the Common Crawl corpus.

Test module	Tests passed
Topic fairness analysis	5
Language fairness analysis	4
PII detection and redaction	7
Toxicity filtering	4
Total	20

The complete workflow is invoked with `python src/run_pipeline.py` after dataset download and extraction. The toxicity stage uses a JSONL checkpoint for resumption. Reproduction requires compatible Python, Java, Spark, and Python dependencies, plus access to the pretrained model.

7. Responsible AI Considerations and Limitations

Quality-filter language effects. The ASCII alphabetic-ratio rule may exclude non-Latin-script documents. Multilingual character-category rules or language-aware quality criteria should be investigated.

Model screening, not adjudication. Toxic-BERT predictions are not ground truth. Automatic removal can exclude legitimate discussion, quotations, reclaimed language, or context-dependent content. Human review and threshold validation are needed for consequential use.

Incomplete PII coverage. Regex-based email and phone detection does not reliably identify all personal information, including names, addresses, identifiers, and obfuscated values. Broader privacy-reviewed testing is needed.

Limited fairness evidence. Automatic language labels are not dialect labels, and observed removal-rate differences do not prove bias. Topic-wise and language-variety evaluations require validated annotations and adequate group sample sizes.

Dataset representativeness. The experiment uses 10,000 documents from one Common Crawl WET segment. Results cannot be generalized to all Common Crawl data or the web. Common Crawl availability also does not imply a uniform licence for every page.

Scalability. Spark ran in local mode with two worker threads. The measured throughput and runtime are environment-specific and do not establish distributed-cluster scalability.

8. Conclusion and Future Work

This project demonstrates a prototype for responsible web-text curation, combining Spark-based ingestion and filtering with PII redaction and transformer-based toxicity screening. From the initial 10,000-document sample, 9,833 remained after exact deduplication, 6,996 after quality filtering, and 6,879 after toxicity screening. PII redaction preserved the record count while detecting email- and phone-like patterns.

Toxic-BERT adds content screening but becomes the computational bottleneck at approximately 2.32 documents per second in the measured environment. The results also highlight potential language-related effects from the ASCII quality rule and substantial differences in observed language-wise toxicity-removal rates. These findings motivate further evaluation but are insufficient to claim model discrimination.

Future work should include: (1) multilingual and language-aware quality filtering; (2) a manually labelled, stratified toxicity evaluation set; (3) validated topic and language-variety annotations; (4) group-wise false-positive and false-negative analysis; (5) threshold and inference optimization; and (6) benchmarking on larger datasets and distributed Spark infrastructure.

References and Project Artifacts

1. Common Crawl, CC-MAIN-2026-43: https://commoncrawl.org/.
2. Toxicity model: `unitary/toxic-bert`, distributed through Hugging Face Transformers.
3. Apache Spark 3.5.6 documentation: https://spark.apache.org/docs/3.5.6/.
4. Implementation: `src/`; experiment outputs: `results/`; results notebook: `notebooks/results_notebook.ipynb`; model and data documentation: `docs/model_card.md` and `docs/data_card.md`.

Reporting note: The report uses recorded experimental metrics. Timings may differ on reruns and other hardware. Topic-wise fairness and manually labelled language-variety evaluation are explicitly reported as pending.